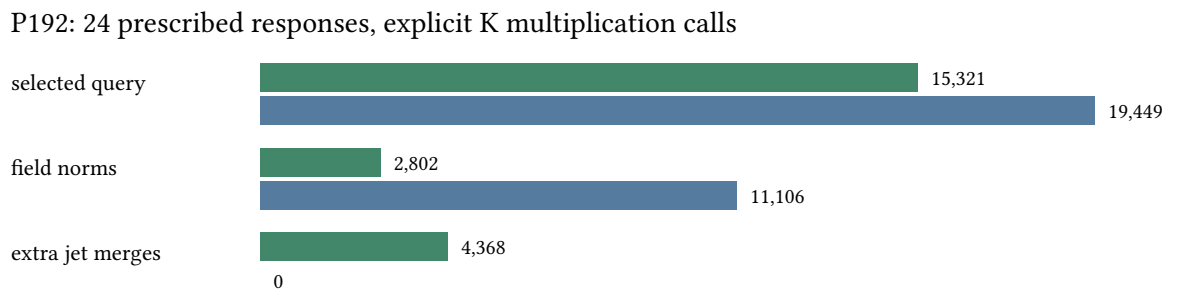
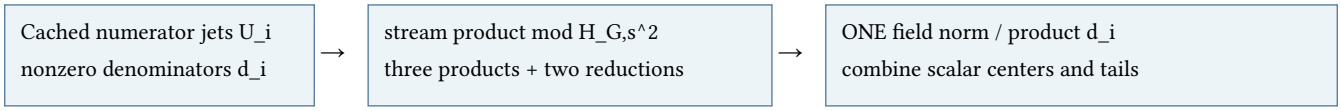
One first norm after giant-block contraction



Green one norm; blue per-block. Shared scale. Constructors unchanged; all merge/reduction costs charged.



Product proof includes zero jets and repeated roots; no numerator norm is inverted.

Round69, 2026-11-09, T-2014. Derived and implemented contraction of complete giant-block numerator jets inside the fixed baby algebra before applying one field norm. Scalar centers and clipped tails remain outside that algebra product. All response and singular controls pass, including repeated zero factors at the same baby root. Scientific standing remains unverified producer evidence. [pcc:unverified]

Construction. In A=K[T]/H_G write U_i=M0_i+s*M1_i for the unchanged Round68 cached numerator and d_i=H_G(x(A_i))^2 H_Q(x(B_i))^2 for its nonzero scalar denominator. Norm multiplicativity gives product_i Norm(U_i)=Norm(product_i U_i). Reduce after each streaming merge, using(A0,A1)*(B0,B1)=(A0B0,A0B1+A1B0) modulo H_G. The first block initializes the accumulator. With scalar center/tail jet C, the result is C*Norm(product U_i)/(product d_i). The written root-product proof in one-norm-proof.md applies over dual numbers and never inverts a zero numerator. [pcc:unverified]

Cost contract. For n complete blocks, replace n first-norm calls by one, plus3(n-1) polynomial products and2(n-1) reductions. Two output divisions occur per query with complete blocks. Denominator evaluations, their scalar product, cache formation, point additions and center/tail work remain charged. Extra accumulator storage is at most2b coefficients, excluding polynomial temporaries and retained reference/qualifier data. Classical polynomial arithmetic remains in use; this contraction does not assign a faster modular-product bound. [pcc:unverified]

Singular qualification. The26 synthetic cases span empty/unit products, a simple zero, two zeros at one root, distinct-root zeros, a zero jet and zero perturbation. Norm of the product matches the product of individual norms, dense determinants and direct root products. Same-root double zeros give a vanishing derivative even with a linear gcd; the simple-root formula evaluates that derivative and returns zero correctly. Wrong multiplication of first coefficients is rejected in10 retained controls. All controls and negative outcomes remain in batch-controls.json. [pcc:unverified]

Frozen response panel. All381 small-curve decoder cases and2694 coefficient queries preserve previous B values, recovered points and decoder counts. The preserved per-block mode's science, cost receipt and P192 output are byte-identical to sealed Round68 records. Candidate small scientific output is byte-identical to that baseline. There are2694 final small-field norm calls instead of11382:924 algebra units and1770 simple-zero outcomes, matching the pre-data predictions. Per-query reference block coefficients are explicitly labeled as reference outputs; selected contraction computes the combined response. [pcc:unverified]

Primary P192. All24 prescribed27-factor responses preserve the earlier publicQ,mu_G,relation_lambda_G coefficient vectors, while72 baby numerator jets are formed. Final norm calls drop72 to24, with6 unit and18 simple outcomes as predicted. Six planted zeros occur in scalar centers. Selected query counts are15321 multiplication calls,645 inversions,168 point-add calls versus19449/1317/168 for per-block norms. Norm work is2802/336 versus11106/1008. Newly charged quotient-jet merges cost4368/96; denominator evaluation/ product912/0; final divisions48/48. Cached formation is unchanged at6408/0. P192 counts exclude inverse internals. These are prescribed windows, not a complete order-r response evaluation. [pcc:unverified]

Small selected-query counts (multiplications including inverse powers,inversions,point-add calls), candidate versus per-block: r7(56820,2200,1400)/(66220,2800,1400); r19(1594168,55544,19296)/(2158609,85475,19296); r199(4093839,75469,10150)/ (8109124,188871,10150). Maximum accumulator slots are2,4,14 respectively, exactly2b. P192 uses8 slots atb4. Source/target point,algebra,H_Q and coefficient-cache constructors are unchanged and separately recorded; no extra target cache is introduced. Counter models remain separate, with no timing or population inference. [pcc:unverified]

Validation and custody. Frozen scope and native source5ad74891641549c075c47b0fab40d7e5b14b19ef precede data. Five science records and two operation receipts reproduce byte-for-byte from a fresh source copy. Changed calibration and P192 coefficient records reject101. No native build failure,timeout or retry occurred. The inherited legacy library compilation timeout124 remains without retry and is not a current-head pass. Publication reconciles advancing main and preserves older artifact-bearing subtree IDs and both ledger prefixes. No cloud job,generator,role,quota or old freeze lane changed. [pcc:unverified]

Next obligation. The remaining contraction is now the modular jet product itself:3(n-1) products and2(n-1) reductions, plus O(nb) formation/denominator work. Investigate whether its fixed biquadratic coefficient family and elliptic giant progression permit a structured evaluation, with explicit assumptions and total setup. This is the next algebraic question rather than an unimplemented fast-product bound. [pcc:unverified]

Sources checked. Round68 coefficient-cache-proof.md, Round64 resultant-proof.md; Stacks0BIE, https://stacks.math.columbia.edu/tag/0BIE, read2026-11-09, for determinant norm/multiplicativity. Bounded study report/proof searches were made; KB tools/wrapper unavailable. No novelty assertion for multiplicativity itself. [pcc:unverified]